THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA REGION TERMINAL EXAMINATION

(F.III) BASIC MATHEMATICS - JUNE 2021

MARKING SCHEME.

SECTION A. (60 MARKS)

01 (a)

2	90	15	20
2	45	15	10
3	45	15	5
3	15	5	5
5	5	5	5
	1	1	1

------- 01 Mark

The LCM $= 2\times2\times3\times3\times5 = 180$ ---- 01 Mark

∴ The smallest number of sweets that can be put into bags is 180 sweets. --- 01 mark

(b) Let $x = 0.9\dot{3}$ ---(i)

multiply by 10 both sides

$10x = 9.3\dot{3}$ ---(ii) ---- 01 Mark

Subtract eqn(i) from (ii)

$10x = 9.3\dot{3}$

$-\ x = 0.9\dot{3}$

$9x = 8.4$ ----- 01 Mark.

$\frac{9x}{9} = \frac{8.4}{9}$

$x = \frac{8.4}{9} = \frac{84}{90} = \frac{42}{45} = \frac{14}{15}$

$\therefore 0.9\dot{3} = \frac{14}{15}$ ---- 01 Mark

02 (a) $\frac{9^{2n}}{3^n} = \frac{1}{27}$

$\frac{3^{4n}}{3^n} = \frac{1}{3^3}$

$3^{4n-n} = 3^{-3}$ ---- 01 Mark.

$4n - n = -3$

$3n = -3$

$n = \frac{-3}{3}$

$\therefore n = -1$ --- 01 Mark

2(b) $\log 64^{\frac{1}{3}} - \log y^2 = \log 64$

$\log (4^3)^{\frac{1}{3}} - \log y^2 = \log 64$ --- 00½ mark

$\log 4 - \log y^2 = \log 64$

$\log 4 - \log 64 = \log y^2$

$\log (4/64) = \log y^2$ ---- 00½ mark.

$4/64 = y^2$

$y^2 = \frac{1}{16}$

$y = \sqrt{1/16}$

$y = 1/4$ ------ 01 mark

2(c) $\left(\frac{2\sqrt{5}+3\sqrt{2}}{2\sqrt{6}-3\sqrt{7}}\right)\left(\frac{2\sqrt{6}+3\sqrt{7}}{2\sqrt{6}+3\sqrt{7}}\right)$ --- 00½ mark

$= \frac{4\sqrt{30}+6\sqrt{35}+6\sqrt{12}+9\sqrt{14}}{4\times 6+6\sqrt{42}-6\sqrt{42}-9\times 7}$

$= \frac{4\sqrt{30}+6\sqrt{35}+6\sqrt{12}+9\sqrt{14}}{24-63}$ --- 00½ mark

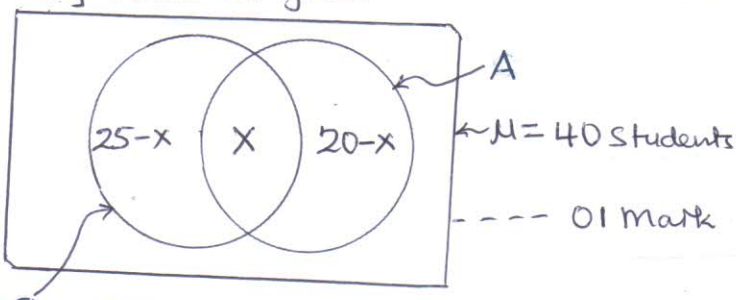
$= \frac{4\sqrt{30}+6\sqrt{35}+6\sqrt{12}+9\sqrt{14}}{-39}$ ---- 01 mark.

3(a) Let C — Number of students who take Commerce

Let A — Number of students who take Agriculture.

Let x — Number of students who take both subjects

By Venn diagram



μ = 40 students

---- 01 mark

$25-x+x+20-x = 40$ ---- 01 mark

$45-x = 40$

$45-40 = x$

$5 = x$

$\therefore x = 5$

$\therefore$ 5 students take both subjects --- 01 mark.

(b) The number of students who take Commerce only

$25-5 = 20$ students --- 01½ marks

3© The number of students who take only one subject are

$20 + 15 = 35$ students --- $01\frac{1}{2}$ marks

4@ $(6,2)$ and $(0,-2)$

Slope $(m) = \frac{y_2 - y_1}{x_2 - x_1}$ or $\frac{y_1 - y_2}{x_1 - x_2}$

Slope $(m) = \frac{-2-2}{0-6} = \frac{-4}{-6} = \frac{2}{3}$ ---- 01 mark

Equation

$\frac{2}{3} = \frac{y+2}{x-0}$

$3(y+2) = 2x$

$3y + 6 = 2x$

$3y = 2x - 6$

$y = \frac{2}{3}x - \frac{6}{3}$

$y = \frac{2}{3}x - 2$ ---- 01 mark

(b) X - intercept when $y = 0$ --- $00\frac{1}{2}$ mark

$0 = \frac{2}{3}x - 2$

$2 = \frac{2}{3}x$ ---- $00\frac{1}{2}$ mark

$6 = 2x$

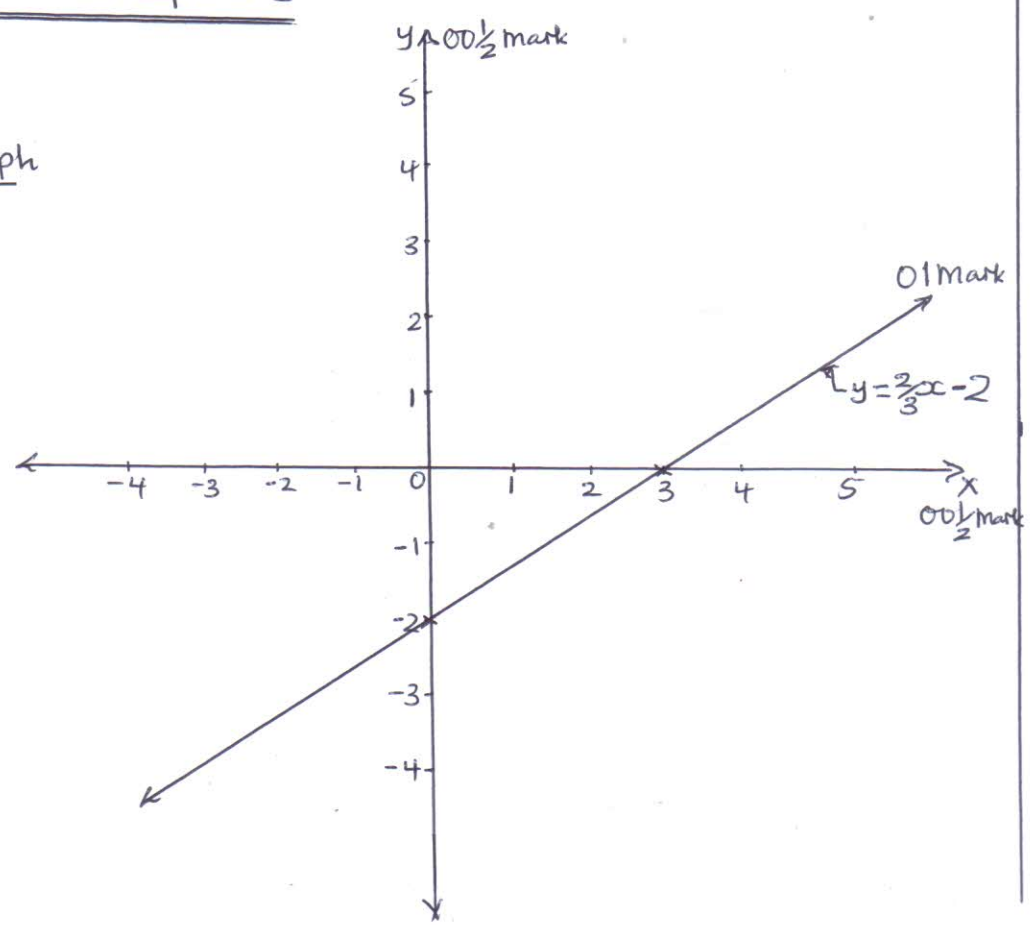
$\frac{6}{2} = x$

$\therefore x = 3 \therefore$ X - intercept $= 3$ --- 01 mark

(c) $y = \frac{2}{3}x - 2$

x	0	3
y	-2	0

Graph



5@(i) Exterior angle + Interior angle $= 180^\circ$ --- 01 mark. 04

Interior angle $= 180^\circ -$ Exterior angle

Interior angle $= 180^\circ - 36^\circ = \underline{144^\circ}$ --- 01 mark.

(ii) Number of Sides $(n) = \dfrac{360^\circ}{\text{Exterior angle}}$

$= \dfrac{360^\circ}{36}$

$= \underline{10 \text{ sides}}$ --- 01 mark.

(b) Area of Trapezium $= \frac{1}{2}(b_1 + b_2)h$

$A = \frac{1}{2}(b_1 + b_2)h$ ---- 01 mark

$121 = \frac{1}{2}(9 + b_2)11$

$121 \times 2 = (9 + b_2)11$

$\dfrac{121 \times 2}{11} = 9 + b_2$

$11 \times 2 = 9 + b_2$ ------ 01 mark.

$22 = 9 + b_2$

$b_2 = 22 - 9$

$b_2 = 13$

The other side is 13cm --- 01 mark.

6@ $x + 5 \propto \dfrac{y^2}{2z}$ ---- 00½ mark.

$x + 5 = \dfrac{ky^2}{2z}$; $x = 5, y = 2$ and $z = 4$

$5 + 5 = \dfrac{4k}{2 \times 4}$ ---- 00½ mark

$10 = \dfrac{4k}{8}$

$k = \dfrac{10 \times 8}{4}$

$\underline{k = 20}$ ----- 01 mark

$x + 5 = \dfrac{20y^2}{2z}$ but $x = 10$ $y = 3$.

$10 + 5 = \dfrac{20 \times 3^2}{2z}$ --- 01 mark

$15 = \dfrac{20 \times 9}{2z}$

$z = \dfrac{20 \times 9}{15 \times 2} = \dfrac{180}{30} = 6$

$\underline{z = 6}$ --- 01 mark.

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$121 \times 2 = (9 + b_2)11$

$\dfrac{121 \times 2}{11} = 9 + b_2$

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$22 = 9 + b_2$

$b_2 = 22 - 9$

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$\underline{k = 20}$ ---- 01 mark

$x + 5 = \dfrac{20y^2}{2z}$ but $x = 10$ $y = 3$.

$10 + 5 = \dfrac{20 \times 3^2}{2z}$ --- 01 mark

$15 = \dfrac{20 \times 9}{2z}$

$z = \dfrac{20 \times 9}{15 \times 2} = \dfrac{180}{30} = 6$

$\underline{z = 6}$ --- 01 mark.

6(b) 10.6m = Total length of a wire

0.53cm = The length of one piece of wire

1m = 100cm

10.6m = 10.6 × 100cm

= 1060cm --- 01 mark

Number of pieces = $\frac{1060cm}{0.53cm} = \frac{106000}{53} = 2000$ pieces -- 01 mark.

7(a) Buying price = 18,000/=

Percentage loss = 15%

Selling price ?

$\%loss = \frac{\text{Buying Price} - \text{Selling price}}{\text{Buying price}} \times 100\%$ --- 01 mark.

$15\% = \frac{18000 - S.P}{18000} \times 100\%$ ----- 01 mark.

$\frac{15}{100} = \frac{18000 - S.P}{18000}$

$0.15 \times 18000 = 18000 - S.P$

$2700 = 18000 - S.P$

$S.P = 18000 - 2700$

$S.P = 15300/=$

∴ Selling price = 15300/= ---- 01 mark.

7(b) $I = \frac{PRT}{100}$ Given $I = 8000/=$ $T = 6yrs$ $R = 15\%$ $P = ?$

$P = \frac{100I}{RT}$ ---- 01 mark

$P = \frac{100 \times 8000}{6 \times 15} = \frac{800000}{90} = 8888.89/=$ ------ 01 mark.

The principal = 8888.89/= ---- 01 mark

8(a) $G_3 = 16$, $G_7 = 1$

$G_1r^2 = 16$ --(i)

$G_1r^6 = 1$ --(ii) ---- 00½ mark

eqn(i) divide by eqn(ii)

$\frac{G_1r^2}{G_1r^6} = \frac{16}{1}$

$\frac{r^2}{r^6} = \frac{16}{1}$

$r^{-4} = 2^4$

$\left(\frac{1}{r}\right)^4 = 2^4$ ∴ $\frac{1}{r} = 2 \Rightarrow r = \frac{1}{2}$ --- 01 mark

8(a) from eqn (i)

$G_1 r^2 = 16$ but $r = \frac{1}{2}$

$G_1(\frac{1}{2})^2 = 16$

$G_1(\frac{1}{4}) = 16$

$G_1 = 16 \times 4 = 64$

$\therefore G_1 = 64$ --- 00½ Mark

$G_n = G_1 r^{n-1}$

$G_{10} = 64 \times (\frac{1}{2})^9$

$G_{10} = 2^6 \times 2^{-9}$

$G_{10} = 2^{-3} = \frac{1}{8}$ --- 01 Mark.

8(b) $S_{13} = 286 \quad d = 3$

$S_n = \frac{n}{2}(2A_1 + (n-1)d)$ ---- 01 Mark

$S_{13} = \frac{13}{2}(2A_1 + 12d)$ but $S_{13} = 286$ and $d = 3$

$286 = \frac{13}{2}(2A_1 + 12 \times 3)$ --- 01 Mark

$\frac{286 \times 2}{13} = 2A_1 + 36$

$44 = 2A_1 + 36$

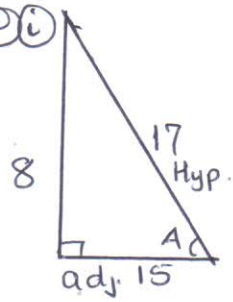
$2A_1 = 44 - 36$

$2A_1 = 8$

$A_1 = 8/2$

$A_1 = 4$ $\therefore$ The first term is 4 --- 01 Mark.

9(a)(i)



By pythagoras theorem

$(Opp)^2 = (Hyp)^2 - (adj)^2$

$(Opp)^2 = (17)^2 - (15)^2$

$Opp = \sqrt{(17+15)(17-15)}$

$= \sqrt{32 \times 2}$

$= \sqrt{64}$

$Opp = 8$ ---- 01 Mark

(i) $\tan A = \frac{Opp}{Adj} = \frac{8}{15}$ --- 01 Mark.

(ii) $\sin A = \frac{Opp}{hyp} = \frac{8}{17}$ --- 01 Mark

9(b) $\overline{PT}^2 = \overline{PQ}^2 - \overline{QT}^2$

$\overline{PT}^2 = (15)^2 - (9)^2$ ---- 00½ Mark.

$\overline{PT}^2 = 225 - 81$

$\overline{PT}^2 = 144$

$PT = \sqrt{144} = 12cm$ $\therefore$ $\overline{PT} = 12cm$ -- 01 Mark

9(b) $\overline{RT}^2 = \overline{PR}^2 - \overline{PT}^2$ ---- 00½ mark.

$= (13)^2 - (12)^2$

$= 169 - 144$

$= 25$

$\overline{RT} = \sqrt{25} = 5 \therefore \overline{RT} = 5cm$ --- 01 mark

10(a) For perfect square;

$b^2 = 4ac$ but $b=1, a=1, c=?$

$(1)^2 = 4(1)(c)$ ----01mark

$1 = 4c$

$c = 1/4 \therefore$ The value of c is $\frac{1}{4}$ ---01mark

(b) $2x^2+7x+6=0$ by Completing the square.

$\frac{2x^2}{2} + \frac{7x}{2} + \frac{6}{2} = \frac{0}{2}$

$x^2 + \frac{7}{2}x + 3 = 0$ -- 01mark.

$x^2 + \frac{7}{2}x = -3$

$x^2 + \frac{7}{2}x + \frac{49}{16} = \frac{49}{16} - 3$ ---01mark

$x^2 + \frac{7}{2}x + \frac{49}{16} = \frac{49-48}{16}$

$x^2 + \frac{7}{2}x + \frac{49}{16} = \frac{1}{16}$

$(x + \frac{7}{4})^2 = \frac{1}{16}$

$x + \frac{7}{4} = \pm\sqrt{\frac{1}{16}}$

$x + \frac{7}{4} = \pm\frac{1}{4}$

$x = -\frac{7}{4} \pm \frac{1}{4}$ ---- 01mark.

Either $x = -\frac{7+1}{4}$ OR $-\frac{7-1}{4}$

$x = -\frac{6}{4}$ OR $-\frac{8}{4}$

$x = -\frac{3}{2}$ OR -2 --- 01mark.

SECTION B (40 MARKS)

11 @ $7+11+20+x+2=45$

$40+x=45$

$x=45-40$

$x=5$ --- 01 mark.

(b)(i) FREQUENCY DISTRIBUTION TABLE.

CLASS INTERVAL	CLASS MARK (x)	FREQUENCY (f)	C.f	fx
45-55	50	7	7	350
56-66	61	11	18	671
67-77	72	20	38	1,440
78-88	83	5	43	415
89-99	94	2	45	188
		N=45		$\Sigma fx = 3064$

--- 02 Marks

$\text{Mean}(\bar{x}) = \frac{\Sigma fx}{N}$

$\bar{x} = \frac{3064}{45}$

Mean $(\bar{x}) = 68.1$ --- 01 mark.

(ii) $\text{Mode} = L + \left(\frac{t_1}{t_1+t_2}\right)i$ --- 01 mark.

Modal Class = 67-77

$L = 66.5,\ t_1 = 9\ \ t_2 = 15\ \ i = 11$

$\text{Mode} = 66.5 + \left(\frac{9}{9+15}\right)11$ --- 01 mark

$= 66.5 + \frac{99}{24}$

$= 66.5 + 4.1$

$= 70.6$ $\therefore$ Mode = 70.6 --- 01 mark.

(iii) $\text{Median} = L + \left(\frac{\frac{N}{2} - n_b}{n_w}\right)i$ --- 01 mark.

Median Class = 67-77

$L = 66.5\ \ \frac{N}{2} = \frac{45}{2} = 22.5,\ n_b = 18$

$n_w = 20\ \ i = 11$

$\text{Median} = 66.5 + \left(\frac{22.5-18}{20}\right)11$ --- 01 mark.

$= 66.5 + \left(\frac{4.5}{20}\right)11$

$= 66.5 + 2.5 = 69$

Median = 69 --- 01 mark.

12@Given 2nd, 4th and 8th = G_1, G_2 and G_3

A_1+d, A_1+3d and A_1+7d = G_1, G_1r and G_1r^2

Common ratio (r)

$\frac{A_1+3d}{A_1+d} = \frac{A_1+7d}{A_1+3d}$ ---- 01 mark

$(A_1+d)(A_1+7d) = (A_1+3d)(A_1+3d)$

$A_1^2 + 8A_1d + 7d^2 = A_1^2 + 6A_1d + 9d^2$

$8A_1d - 6A_1d = 9d^2 - 7d^2$

$2A_1d = 2d^2$ ---- 01 mark

$\frac{2A_1d}{2d} = \frac{2d^2}{2d}$

$A_1 = d$ ---(i)--- 01 mark

$3^{rd} + 5^{th} = 20$

$A_1+2d + A_1+4d = 20$

$2A_1+6d = 20$ ---(ii)--- 01 mark

but $A_1 = d$

$2d+6d = 20$

$8d = 20$

$d = \frac{20}{8}$

$d = 5/2$

Common difference (d) = $5/2$ ---01 mark

(b) Common ratio (r) = $\frac{A_1+3d}{A_1+d}$ but $d = A_1$

$r = \frac{d+3d}{d+d}$ ---- 01 mark

$r = \frac{4d}{2d}$

$r = 2$

Common ratio (r) = 2 ---01 mark

(c) $S_n = \frac{n}{2}(2A_1 + (n-1)d$ ---- 01 mark

but $n = 9$ $A_1 = 5/2$ and $d = 5/2$

$S_9 = \frac{9}{2}(2\times\frac{5}{2} + (9-1)\frac{5}{2})$ ---- 01 mark

$S_9 = 4.5(5 + 8\times 5/2)$

$S_9 = 4.5(5+20)$

$S_9 = 4.5(25)$

$S_9 = 112.5$

∴ The sum of the first 9 terms of an AP = 112.5 --- 01 mark

13. (a) (i) T $(0,0)$ ——— $(-3,-7)$

$(x', y') = (x, y) + (a, b)$ ---- 00½ mark.

$(-3,-7) = (0,0) + (a,b)$

$(a,b) = (-3,-7)$

$\therefore (x', y') = (2,6) + (-3,-7) = (-1,-1)$

$\therefore T(2,6) \longrightarrow (-1,-1)$ ---01 mark

(ii) $M_{y=x}(x,y) = (y,x)$ --- 00½ mark

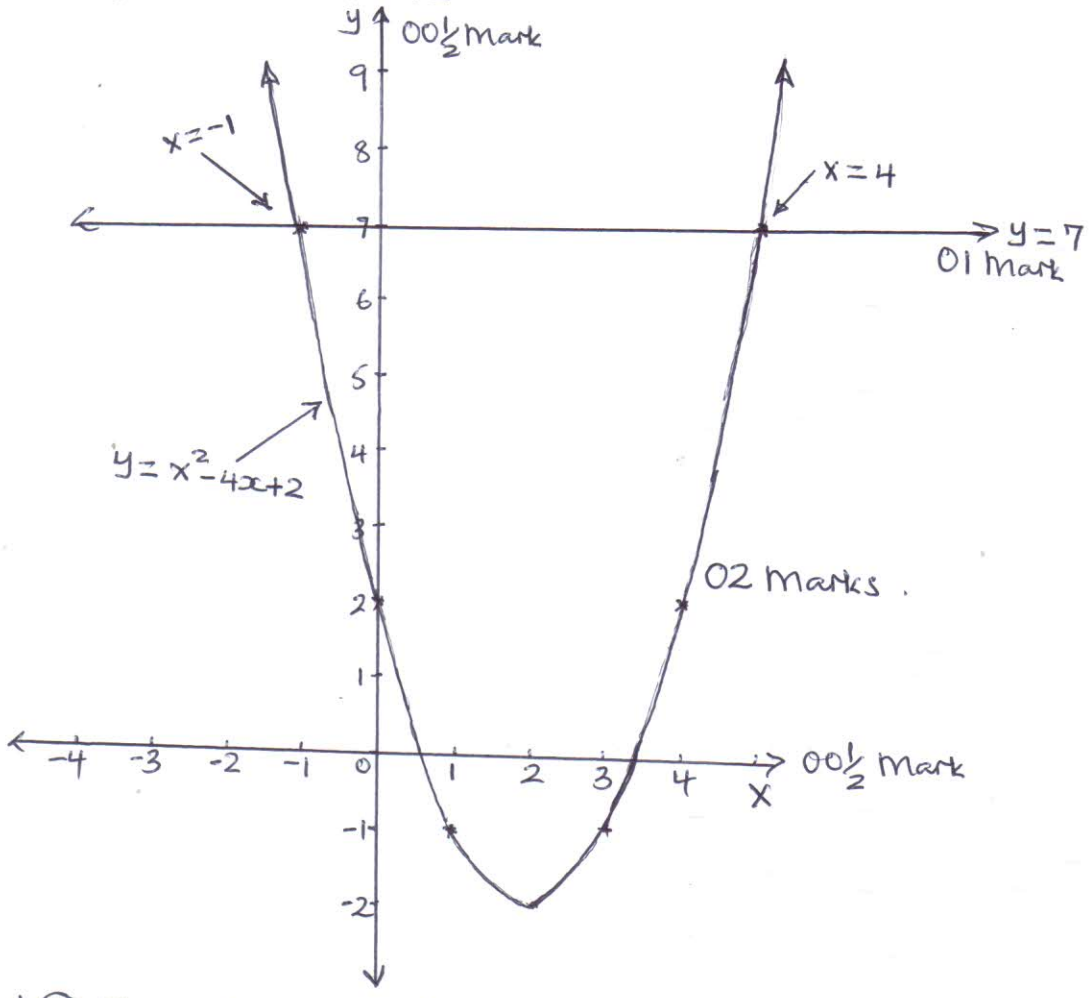
$M_{y=x}(-4,3) = (3,-4)$ --- 01 mark.

(b) Table of values for $y = x^4 - 4x + 2$

x	-2	-1	0	1	2	3	4	5	6
y	10	7	2	-1	-2	-1	2	7	10

01 mark

Graph of $y = x^2 - 4x + 2$.



(b) (i) From the graph the minimum value of y is -2 and the corresponding value of x is 2 ---01 mark.

(ii) Compare $y = x^2 - 4x + 2$ and $x^2 - 4x - 2 = 3$

$x^2 - 4x + 2 = y$ and $x^2 - 4x - 2 - 3 = 0$

$x^2 - 4x + 2 = y$ and $x^2 - 4x - 5 = 0$

$x^2 - 4x + 2 = y$ and $x^2 - 4x + 2 - 7 = 0$

$x^2 - 4x + 2 = y$ and $x^2 - 4x + 2 = 7$

Then $y = 7$

From the graph above the values of x are -1 or 4 01 mark.

14@ Domain $= \{0, 1, 3\}$ 02 Marks

Range $= \{2, 5\}$ 02 marks

(b)(i) $y = |x|$ if $-3 \leq x \leq 3$

x	-3	-2	-1	1	2	3
y	3	2	1	1	2	3

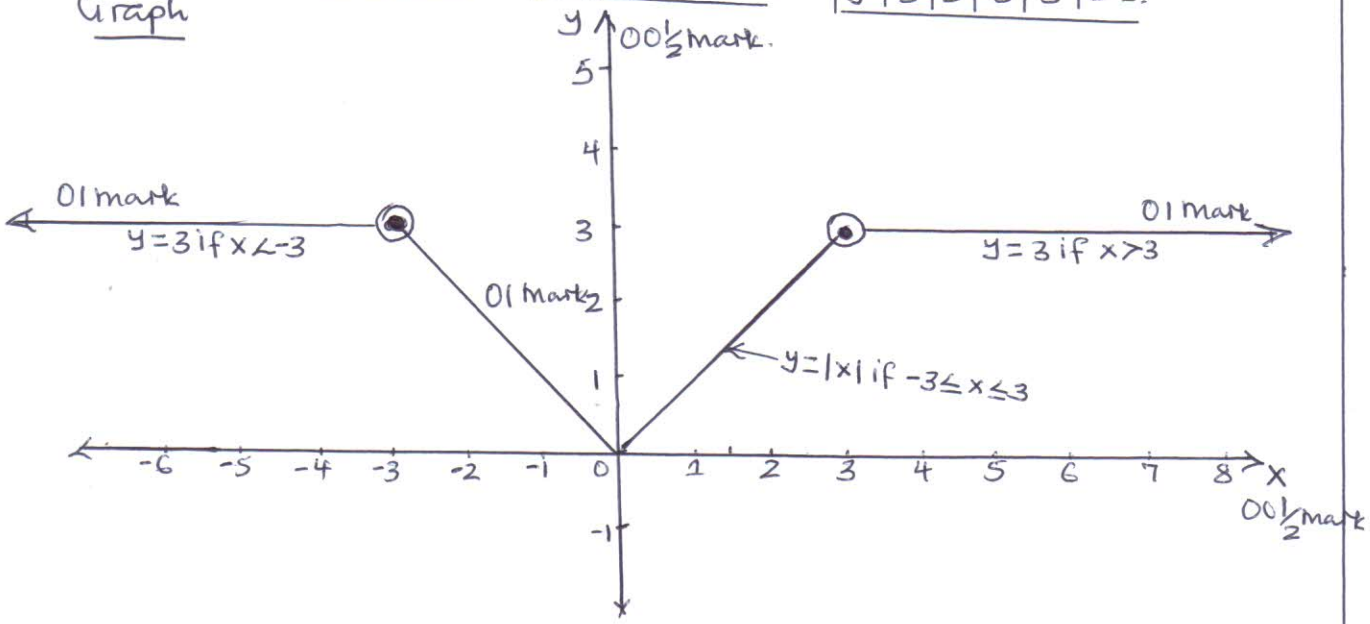
$y = 3$ if $x > 3$

x	3	4	5	--
y	3	3	3	--

$y = 3$ if $x < -3$

x	-3	-4	-5	-6	--
y	3	3	3	3	--

Graph



(ii) Domain $= \{x : x$ is all real numbers$\}$ 01 mark

Range $= \{y : 0 \leq y \leq 3\}$ 01 mark.

THE END